bytes that a datagram carries, when all 16 bits are 1s, is 65,535. IHL field counts only the header fields in units of 32-bit or 4-byte words. But TL counts the entire packets in bytes.

The entire IP datagram is variable and so are header fields. Using IHL the receiver identifies the number of header bytes in a packet. The receiver, by subtracting header bytes from total bytes, identifies the beginning and end of data, which is the TCP pack.

At the second level, there are three headers, as follows:

• In the identification field, a 16-bit header connects the different types of networks designed to carry packets of different sizes. For example, Ethernet packets may be in the rage of 64 to 1518 bytes, MAN may have a maximum packet size of 1500 bytes, FDDI packet can be of 4472 bytes and a token ring packet may be typically of 17,800 bytes. When a packet of, say, 1500 bytes arrives at a node or router, and the same is required to be transferred or routed to a network, say, having maximum packet size of 512 bytes, it is required that the arriving packet of 1500 bytes is fragmented into several smaller packets, each having a size of 512 bytes or less.

Each part or fragmented packet of a particular original packet must be given some identification so that the receiver can recombine all related fragments in packets. Same identification to all fragments of a packet ensures that wrong fragments are not combined to reconstruct the original packet in the receiver. The identification field assigns a single group number to all fragments of a packet being fragmented.

• 3-bit flags are used to control and convey information with regards to fragmentation. The left-most bit is reserved. The middle bit known as don't fragment (DF) provides flexibility to the sender. If the sender wishes that the packet must not be fragmented, even if it does not reach the destination, he or she can construct an IP datagram with DF bit set to 1.

When DF bit is set to zero, fragmentation is allowed. When a node or router breaks a packet into several fragments, more fragment (MF) bit of all fragments, except the last one, is set to 1. MF bit is set to 0 in the last fragment. By this, the receiver is made aware about the last fragment of a packet.

• 13-bit fragment offset indicates how to insert fragments in the receiver's buffer to reconstruct the original datagram. The field is measured in units of 8 bytes. This is because there are only 2^{13} offset values but the datagram can be as big as 2^{16} bytes. So each fragment must be in multiples of 8 bytes, as $2^{13}/2^{16} = 8$.

This requirement applies only to the data field (or TCP pack) of the original packet, because header fields of the original IP by right are also the header fields of each fragment. One cannot expect the header fields to be always divisible by 8. So a fragment's data field will always be in multiples of 8 bytes, but total length of the fragment (total length of fragment = original packet header length + data length of fragment) will be in multiples of 4 bytes.

However, total length of the fragment may be made in multiples of 8 bytes by padding for the purpose of storing at the receiver. Multiples of 8 bytes are only applicable to data fields, not to the whole datagram. In the worst case, data fields may be just 8 bytes. So for permissible header lengths, the worst-case fragmented packet sizes would be 28, 32, 36, 40, 44, 48, 52, 56, 60, 64 and 68 bytes.

But IP does not run over the networks that have the maximum transmission unit (MTU) smaller than 68 bytes—this is the highest fragment size as a worst-case scenario. Thus, for an IP datagram of 20 bytes IHL, the worst case refers to fragments having data field in multiples of 48 (68-20) bytes.

Since, 48 bytes are multiples of 8 bytes, consider an original IP packet with 20 bytes minimum IHL and 65515 (65535-20) bytes maximum TL. In the worst case, an original IP packet may be fragmented into 1364 (65515/48) fragments with remaining 43 bytes left over as the 1365th fragment, which is made 68 bytes long by padding.

Headers at the third level are:

• 8-bit lifetime or time-to-live (TTL) field that is used to ensure that an IP packet does not persist in the Internet forever. If the packet persists in the network forever, congestion is bound to occur. So TTL, in its own right, provides a mechanism to avoid congestion in the Internet. Maximum TTL field is 255. In the early days, it was taken as 255 seconds. Presently, TTL field is taken as a hop counter. Thus, an IP packet can move maximum 255 hops in the Internet.

The transmitting station sets the TTL field of the datagram. When the

0 1 2 3 4 8 16 24 31

Class A	0	Net ID | Host ID
Class B	1 0	Net ID | Host ID
Class C	1 1 0	Net ID | Host ID
Class D	1 1 1 0	Multicast address
Class E	1 1 1 1 0	Reserved for future use

(a) Structure of IPv4 addressing (netid=network identification, hostid=host identification)

Different classes	Starting address	Last address
Class-A	**0**.0.0.0	**127**.255.255.255
Class-B	**128.0**.0.0	**191.255**.255.255
Class-C	**192.0.0**.0	**223.255.255**.255
Class-D	224.0.0.0	239.255.255.255
Class-E	240.0.0.0	247.255.255.255

(b) Address space under different classes. (Bold space is for netid and rest is for hostid; except in case for class-E. In case of class-E, the whole of the address space id undefined)

Class	Range	Number of networks	Number of hosts per network
A	1.0.0.0 to 126.254.254.254	126	16,000,000
B	128.0.0.0 to 191.254.254.254	16,000	64,000
C	192.0.0.0 to 223.254.254.254	2,000,000	265
D	224.0.0.0 to 239.0.0.0	NA	Multicast
E	240.0.0.0 to 255.0.0.0	NA	Test
	127.0.0.0 to 127.254.254.254	NA	Loop back
	255.255.255.255	NA	Broadcast

(c) Details in different addresses

Fig. 5: Illustration of IPv4 address scheme